# Ultimate AI Chat Interface - Technical Specification

## 🎯 Project Overview

Build a next-generation AI chat interface that combines project management, document processing, and conversational AI in a visually stunning, highly addictive user experience that surpasses existing solutions.

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## 📋 Core Requirements Summary

### Primary Features

- **Project/Agent Management System** - Visual project cards with CRUD operations
- **Advanced Chat Interface** - Real-time messaging with rich media support
- **Document Ingestion System** - Drag-and-drop file processing with progress tracking
- **Dual Theme System** - Sophisticated dark mode (primary) and elegant light mode
- **Progressive Web App** - Mobile-first responsive design with offline capabilities

### Performance Targets

- **Load Time**: < 2 seconds initial page load
- **Animation Performance**: Consistent 60 FPS
- **Accessibility**: WCAG 2.1 AA compliance
- **Mobile Performance**: 90+ Lighthouse scores

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## 🎨 Design System Specification

### Color Palettes

#### Dark Mode (Primary Theme) - "Neo-Futuristic"

```css
:root[data-theme="dark"] {
  /* Background Layers */
  --bg-primary: #0a0a0f;
  --bg-secondary: #141419;
  --bg-tertiary: #1a1a22;
  --bg-glass: rgba(20, 20, 25, 0.8);

  /* Accent Colors */
  --accent-electric: #00d4ff;
  --accent-purple: #8b5cf6;
  --accent-plasma: #00ff88;
  --accent-gold: #ffd700;

  /* Text Colors */
  --text-primary: #ffffff;
  --text-secondary: #a0a0ab;
  --text-muted: #6b6b76;

  /* Interactive States */
  --hover-overlay: rgba(0, 212, 255, 0.1);
  --active-overlay: rgba(0, 212, 255, 0.2);
  --focus-ring: rgba(0, 212, 255, 0.3);

  /* Gradients */
  --gradient-primary: linear-gradient(135deg, #8b5cf6 0%, #00d4ff 100%);
  --gradient-secondary: linear-gradient(135deg, #00ff88 0%, #ffd700 100%);
  --gradient-subtle: linear-gradient(135deg, rgba(139, 92, 246, 0.1) 0%, rgba(0, 212,
}
```

**Light Mode - "Elegant Minimalism"**

```css
:root[data-theme="light"] {
  /* Background Layers */
  --bg-primary: #fafafa;
  --bg-secondary: #ffffff;
  --bg-tertiary: #f5f5f7;
  --bg-glass: rgba(255, 255, 255, 0.9);

  /* Accent Colors */
  --accent-electric: #0071e3;
  --accent-purple: #6b46c1;
  --accent-plasma: #059669;
  --accent-gold: #d97706;

  /* Text Colors */
  --text-primary: #1d1d1f;
  --text-secondary: #515154;
  --text-muted: #86868b;

  /* Interactive States */
  --hover-overlay: rgba(0, 113, 227, 0.08);
  --active-overlay: rgba(0, 113, 227, 0.15);
  --focus-ring: rgba(0, 113, 227, 0.3);

  /* Gradients */
  --gradient-primary: linear-gradient(135deg, #6b46c1 0%, #0071e3 100%);
  --gradient-secondary: linear-gradient(135deg, #059669 0%, #d97706 100%);
  --gradient-subtle: linear-gradient(135deg, rgba(107, 70, 193, 0.05) 0%, rgba(0, 113,
}
```

## Typography System

```css
/* Font Stack */
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&d

:root {
  --font-primary: 'Inter', -apple-system, BlinkMacSystemFont, sans-serif;
  --font-mono: 'SF Mono', Monaco, 'Cascadia Code', monospace;

  /* Type Scale */
  --text-xs: 0.75rem;      /* 12px */
  --text-sm: 0.875rem;     /* 14px */
  --text-base: 1rem;       /* 16px */
  --text-lg: 1.125rem;     /* 18px */
  --text-xl: 1.25rem;      /* 20px */
  --text-2xl: 1.5rem;      /* 24px */
  --text-3xl: 1.875rem;    /* 30px */
  --text-4xl: 2.25rem;     /* 36px */

  /* Line Heights */
  --leading-tight: 1.25;
  --leading-normal: 1.5;
  --leading-relaxed: 1.75;
}
```

## Spacing & Layout

```css
:root {
  /* Spacing Scale */
  --space-1: 0.25rem;   /* 4px */
  --space-2: 0.5rem;    /* 8px */
  --space-3: 0.75rem;   /* 12px */
  --space-4: 1rem;      /* 16px */
  --space-6: 1.5rem;    /* 24px */
  --space-8: 2rem;      /* 32px */
  --space-12: 3rem;     /* 48px */
  --space-16: 4rem;     /* 64px */
  --space-20: 5rem;     /* 80px */

  /* Border Radius */
  --radius-sm: 0.375rem;    /* 6px */
  --radius-md: 0.5rem;      /* 8px */
  --radius-lg: 0.75rem;     /* 12px */
  --radius-xl: 1rem;        /* 16px */
  --radius-2xl: 1.5rem;     /* 24px */
  --radius-full: 9999px;

  /* Shadows */
  --shadow-sm: 0 1px 2px 0 rgba(0, 0, 0, 0.05);
  --shadow-md: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
  --shadow-lg: 0 10px 15px -3px rgba(0, 0, 0, 0.1);
  --shadow-xl: 0 20px 25px -5px rgba(0, 0, 0, 0.1);
  --shadow-glow: 0 0 20px rgba(0, 212, 255, 0.3);
}
```

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## 🏗️ Architecture & File Structure

### Project Structure

```
src/
├── components/
│   ├── ui/                    # Base UI components
│   │   ├── Button/
│   │   ├── Input/
│   │   ├── Modal/
│   │   ├── Card/
│   │   └── index.ts
│   ├── chat/                  # Chat-specific components
│   │   ├── ChatInterface/
│   │   ├── MessageBubble/
│   │   ├── InputArea/
│   │   ├── TypingIndicator/
│   │   └── index.ts
│   ├── project/               # Project management
│   │   ├── ProjectCard/
│   │   ├── ProjectGrid/
│   │   ├── ProjectModal/
│   │   └── index.ts
│   ├── document/              # Document handling
│   │   ├── DocumentUpload/
│   │   ├── DocumentPreview/
│   │   ├── ProcessingStatus/
│   │   └── index.ts
│   └── layout/                # Layout components
│       ├── Sidebar/
│       ├── Header/
│       ├── MainContent/
│       └── index.ts
├── hooks/                     # Custom React hooks
│   ├── useChat.ts
│   ├── useProjects.ts
│   ├── useDocuments.ts
│   ├── useTheme.ts
│   └── useAnimations.ts
├── stores/                    # State management
│   ├── chatStore.ts
│   ├── projectStore.ts
│   ├── documentStore.ts
│   └── themeStore.ts
├── utils/                     # Utility functions
│   ├── animations.ts
│   ├── api.ts
```

```
│   ├── storage.ts
│   └── validation.ts
├── styles/                    # Global styles
│   ├── globals.css
│   ├── animations.css
│   ├── components.css
│   └── themes.css
└── types/                     # TypeScript definitions
    ├── chat.ts
    ├── project.ts
    ├── document.ts
    └── common.ts
```

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## 🎭 Animation System Specification

### Core Animation Principles

- **60 FPS Performance**: Use CSS transforms and opacity only
- **Easing Functions**: Custom cubic-bezier curves for organic motion
- **Staggered Animations**: Sequential reveals for list items
- **Micro-interactions**: Immediate feedback for all user actions

### Animation Library

css

```css
/* Easing Functions */
:root {
  --ease-out-quad: cubic-bezier(0.25, 0.46, 0.45, 0.94);
  --ease-out-quart: cubic-bezier(0.165, 0.84, 0.44, 1);
  --ease-in-out-quint: cubic-bezier(0.86, 0, 0.07, 1);
  --ease-elastic: cubic-bezier(0.68, -0.55, 0.265, 1.55);
}

/* Base Animations */
@keyframes fadeInUp {
  from {
    opacity: 0;
    transform: translateY(20px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@keyframes scaleIn {
  from {
    opacity: 0;
    transform: scale(0.95);
  }
  to {
    opacity: 1;
    transform: scale(1);
  }
}

@keyframes slideInRight {
  from {
    opacity: 0;
    transform: translateX(30px);
  }
  to {
    opacity: 1;
    transform: translateX(0);
  }
}

@keyframes pulse {
```

```
  0%, 100% {
    opacity: 1;
  }
  50% {
    opacity: 0.7;
  }
}

@keyframes float {
  0%, 100% {
    transform: translateY(0px);
  }
  50% {
    transform: translateY(-10px);
  }
}

/* Utility Classes */
.animate-fade-in-up {
  animation: fadeInUp 0.6s var(--ease-out-quart) both;
}

.animate-scale-in {
  animation: scaleIn 0.4s var(--ease-out-quad) both;
}

.animate-slide-in-right {
  animation: slideInRight 0.5s var(--ease-out-quad) both;
}

.animate-pulse {
  animation: pulse 2s infinite;
}

.animate-float {
  animation: float 3s ease-in-out infinite;
}

/* Stagger Delays */
.stagger-1 { animation-delay: 0.1s; }
.stagger-2 { animation-delay: 0.2s; }
.stagger-3 { animation-delay: 0.3s; }
.stagger-4 { animation-delay: 0.4s; }
```

## 🧩 Component Specifications

### 1. Chat Interface (`/components/chat/ChatInterface`)

#### File Structure

```
ChatInterface/
├── ChatInterface.tsx
├── ChatInterface.module.css
├── ChatInterface.test.tsx
└── index.ts
```

#### Component Requirements

```typescript
interface ChatInterfaceProps {
  projectId?: string;
  className?: string;
  height?: string | number;
}

interface Message {
  id: string;
  content: string;
  role: 'user' | 'assistant';
  timestamp: Date;
  status: 'sending' | 'sent' | 'error';
  attachments?: Attachment[];
}
```

#### Key Features

- Real-time message streaming
- Markdown rendering with syntax highlighting
- Message threading and branching
- Voice input/output integration
- Responsive mobile-first design
- Infinite scroll with virtualization

### 2. Message Bubble (`/components/chat/MessageBubble`)

#### Design Specifications

- **User Messages**: Right-aligned, gradient background, rounded corners
- **AI Messages**: Left-aligned, glass morphism effect, typing animation
- **States**: Sending, sent, error with visual indicators
- **Interactions**: Copy, edit, delete, branch conversation

### 3. Input Area (`/components/chat/InputArea`)

#### Features Required

- Auto-expanding textarea with max height
- File attachment with drag-and-drop
- Voice recording with waveform visualization
- Send button with loading states
- Keyboard shortcuts (Cmd/Ctrl + Enter)
- Emoji picker integration

### 4. Project Card (`/components/project/ProjectCard`)

#### Visual Requirements

- Glass morphism background with blur
- Gradient borders on hover
- Project status indicators with colors
- Usage statistics with mini charts
- Context menu for actions
- Drag-and-drop reordering

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## 🎨 Visual Effects Implementation

### Glass Morphism System

```css
.glass-morphism {
  background: var(--bg-glass);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  border: 1px solid rgba(255, 255, 255, 0.1);
  box-shadow:
    0 8px 32px rgba(0, 0, 0, 0.1),
    inset 0 1px 0 rgba(255, 255, 255, 0.1);
}

.glass-morphism-light {
  background: rgba(255, 255, 255, 0.9);
  backdrop-filter: blur(20px);
  border: 1px solid rgba(0, 0, 0, 0.05);
  box-shadow:
    0 8px 32px rgba(0, 0, 0, 0.05),
    inset 0 1px 0 rgba(255, 255, 255, 0.8);
}
```

### Particle System

typescript

```typescript
// Particle system for background effects
interface Particle {
  x: number;
  y: number;
  vx: number;
  vy: number;
  size: number;
  opacity: number;
  color: string;
}

class ParticleSystem {
  private particles: Particle[] = [];
  private canvas: HTMLCanvasElement;
  private ctx: CanvasRenderingContext2D;

  constructor(canvas: HTMLCanvasElement) {
    this.canvas = canvas;
    this.ctx = canvas.getContext('2d')!;
    this.init();
  }

  private init() {
    // Initialize 50-100 particles
    // Gentle floating motion
    // Color based on current theme
  }

  public render() {
    // Render particles with 60 FPS
    // Use requestAnimationFrame
  }
}
```

## Glow Effects

```css
.glow-electric {
  box-shadow: 0 0 20px rgba(0, 212, 255, 0.3);
  transition: box-shadow 0.3s var(--ease-out-quad);
}

.glow-electric:hover {
  box-shadow: 0 0 30px rgba(0, 212, 255, 0.5);
}

.glow-purple {
  box-shadow: 0 0 20px rgba(139, 92, 246, 0.3);
}

.glow-plasma {
  box-shadow: 0 0 20px rgba(0, 255, 136, 0.3);
}
```

---

## 📱 Responsive Design Specification

### Breakpoint System

```css
:root {
  --breakpoint-sm: 640px;
  --breakpoint-md: 768px;
  --breakpoint-lg: 1024px;
  --breakpoint-xl: 1280px;
  --breakpoint-2xl: 1536px;
}
```

### Layout Variations

- **Mobile (< 768px)**: Single column, bottom navigation, swipe gestures
- **Tablet (768px - 1024px)**: Split view, gesture navigation
- **Desktop (> 1024px)**: Multi-panel layout, keyboard shortcuts
- **Large Desktop (> 1280px)**: Extended sidebar, more visual elements

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## 🔧 State Management Architecture

## Chat Store (Zustand)

typescript

```typescript
interface ChatState {
  messages: Message[];
  currentThreadId: string | null;
  isTyping: boolean;

  // Actions
  addMessage: (message: Omit<Message, 'id'>) => void;
  updateMessage: (id: string, updates: Partial<Message>) => void;
  deleteMessage: (id: string) => void;
  setTyping: (isTyping: boolean) => void;
  clearChat: () => void;
}

export const useChatStore = create<ChatState>((set, get) => ({
  messages: [],
  currentThreadId: null,
  isTyping: false,

  addMessage: (message) => set((state) => ({
    messages: [...state.messages, { ...message, id: generateId() }]
  })),

  updateMessage: (id, updates) => set((state) => ({
    messages: state.messages.map(msg =>
      msg.id === id ? { ...msg, ...updates } : msg
    )
  })),

  // ... other actions
}));
```

## Project Store

typescript

```typescript
interface ProjectState {
  projects: Project[];
  activeProject: Project | null;

  // Actions
  createProject: (project: CreateProjectData) => Promise<Project>;
  updateProject: (id: string, updates: Partial<Project>) => Promise<void>;
  deleteProject: (id: string) => Promise<void>;
  setActiveProject: (project: Project | null) => void;
}
```

---

## 🎯 Cursor AI Implementation Rules

### STRICT CURSOR RULES

#### 1. File Creation and Organization

```
RULE: Always create complete file structures before writing code
RULE: Use TypeScript for all React components and utilities
RULE: Create accompanying test files for each component
RULE: Generate CSS modules for component-specific styles
RULE: Export components through index.ts barrel files
```

#### 2. Code Quality Standards

```
RULE: Use functional components with hooks only
RULE: Implement proper TypeScript interfaces for all props
RULE: Include JSDoc comments for complex functions
RULE: Use semantic HTML elements for accessibility
RULE: Implement error boundaries for production robustness
RULE: Use React.memo for expensive components
```

#### 3. Styling Guidelines

```
RULE: Use CSS modules with descriptive class names
RULE: Implement both dark and light theme variants
RULE: Use CSS custom properties for theme variables
RULE: Ensure 60 FPS animations using transform/opacity only
RULE: Include :focus-visible states for keyboard navigation
RULE: Implement glass morphism using backdrop-filter
```

### 4. State Management Rules

```
RULE: Use Zustand for global state management
RULE: Implement React Query for server state
RULE: Use localStorage for theme persistence only
RULE: Implement optimistic updates for better UX
RULE: Handle loading and error states explicitly
```

### 5. Performance Requirements

```
RULE: Implement React.lazy for code splitting
RULE: Use IntersectionObserver for infinite scroll
RULE: Implement virtual scrolling for large lists
RULE: Optimize images with next/image or similar
RULE: Use React.Suspense for async components
RULE: Implement service worker for offline functionality
```

## CURSOR METAPROMPTS

### Component Creation Metaprompt

```
When creating a new component:
1. Generate the folder structure with all required files
2. Create the main component with TypeScript interfaces
3. Implement both dark and light theme styles
4. Add proper accessibility attributes
5. Include loading and error states
6. Write unit tests with React Testing Library
7. Create Storybook stories for design system documentation
8. Export through index.ts with proper TypeScript exports
```

### Animation Implementation Metaprompt

```
When implementing animations:
1. Use CSS transforms and opacity only for 60 FPS performance
2. Implement proper easing functions from the design system
3. Add reduced motion media query support
4. Use Framer Motion for complex orchestrated animations
5. Implement staggered animations for list items
6. Add proper cleanup in useEffect hooks
7. Test on mobile devices for performance validation
```

### State Management Metaprompt

```
When implementing state management:
1. Use Zustand stores with TypeScript interfaces
2. Implement proper action creators with error handling
3. Add optimistic updates for better UX
4. Use React Query for server state with proper caching
5. Implement proper error boundaries and fallbacks
6. Add persistence only where explicitly needed
7. Use proper loading states throughout the application
```

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## 🧪 Testing Strategy

### Unit Testing Requirements

typescript

```typescript
// Example test structure
describe('ChatInterface', () => {
  it('renders messages correctly', () => {
    // Test message rendering
  });

  it('handles user input', () => {
    // Test input handling
  });

  it('supports keyboard shortcuts', () => {
    // Test accessibility
  });

  it('handles error states gracefully', () => {
    // Test error handling
  });
});
```

### Performance Testing

- Lighthouse CI integration
- Core Web Vitals monitoring
- Animation frame rate testing
- Memory leak detection

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## 🚀 Deployment & CI/CD

### Build Configuration

```json
{
  "scripts": {
    "dev": "next dev",
    "build": "next build && next export",
    "start": "next start",
    "lint": "eslint . --ext .ts,.tsx",
    "test": "jest",
    "test:e2e": "playwright test",
    "analyze": "ANALYZE=true next build"
  }
}
```

### Environment Variables

```env
NEXT_PUBLIC_API_URL=
NEXT_PUBLIC_WEBSOCKET_URL=
NEXT_PUBLIC_ANALYTICS_ID=
DATABASE_URL=
OPENAI_API_KEY=
```

---

## 📊 Analytics & Monitoring

### User Experience Metrics

- Time to first interaction
- Session duration
- Feature usage patterns
- Error rate tracking
- Performance metrics

### Implementation

```typescript
// Analytics tracking
interface AnalyticsEvent {
  event: string;
  properties: Record<string, any>;
  timestamp: Date;
}

export const trackEvent = (event: AnalyticsEvent) => {
  // Implementation
};
```

---

## 🎨 Design System Documentation

### Component Library Structure

- Storybook integration for visual documentation
- Design tokens exported as CSS custom properties
- Interactive component playground
- Accessibility testing integrated
- Visual regression testing with Chromatic

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## 🔒 Security Considerations

### Data Protection

- XSS prevention with DOMPurify
- CSP headers implementation
- Secure file upload validation
- Rate limiting for API endpoints
- Input sanitization for all user data

### Privacy

- Local-first architecture where possible
- Minimal data collection
- Clear privacy policy implementation
- GDPR compliance considerations

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## 🎯 Success Metrics & KPIs

### User Engagement

- **Target**: 15+ minute average session
- **Target**: 80%+ return rate within 24 hours
- **Target**: 90%+ task completion rate
- **Target**: < 3 second time to "wow" moment

### Technical Performance

- **Target**: < 2 second initial load time
- **Target**: 60 FPS animation performance
- **Target**: 90+ Lighthouse scores across all categories
- **Target**: < 1% error rate

### Accessibility

- **Target**: WCAG 2.1 AA compliance
- **Target**: 100% keyboard navigation support
- **Target**: Screen reader compatibility
- **Target**: High contrast mode support

---

## 📝 Implementation Checklist

### Phase 1: Foundation (Week 1)

- [ ] Project setup with Next.js and TypeScript
- [ ] Design system implementation
- [ ] Basic layout components
- [ ] Theme switching functionality
- [ ] Core state management setup

### Phase 2: Core Features (Week 2)

- [ ] Chat interface implementation
- [ ] Message bubble components
- [ ] Input area with file upload
- [ ] Project management system

- [ ] Document processing interface

### Phase 3: Advanced Features (Week 3)

- [ ] Real-time messaging with WebSockets
- [ ] Voice input/output integration
- [ ] Advanced animations and micro-interactions
- [ ] Mobile responsive optimization
- [ ] Progressive Web App features

### Phase 4: Polish & Launch (Week 4)

- [ ] Performance optimization
- [ ] Accessibility audit and fixes
- [ ] Cross-browser testing
- [ ] Security audit
- [ ] Production deployment

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## 🎭 Final Implementation Notes

### Code Quality Gates

- All components must pass TypeScript strict mode
- 90%+ test coverage requirement
- Lighthouse performance score > 90
- Zero accessibility violations
- Cross-browser compatibility verified

### Design Validation

- Visual designs must match Figma specifications exactly
- Animations must run at 60 FPS on low-end devices
- Color contrast ratios must meet WCAG standards
- Interactive elements must have proper focus states
- Mobile experience must be touch-optimized

This specification provides the complete technical foundation for building the ultimate AI chat interface. Every detail has been carefully considered to ensure the final product meets the ambitious vision outlined in the original metaprompt.